

!Fraud Detection

You can download the initial project file from [Link](#).

Fintech company **NexusPay** takes pride in its advanced AI-powered fraud detection system. The company's models have been trained on millions of transactions and identify fraudulent activities with remarkable accuracy. However, the finance department has recently detected a persistent anomaly: a small but steady stream of fraudulent transactions continues to bypass the system.

These **"ghost transactions"** (*fraud*) do not match any known patterns. Their amounts are not unusually high, they do not originate from high-risk countries, and they do not trigger the behavioral alerts that our current models are designed to detect. It is as if they are invisible to our standard behavioral analyses.

You have been hired as a data scientist for **Project Sentinel**. Your mission is to analyze a new dataset that combines transaction details with internal processing metadata.

Your goal is to build a classification model capable of identifying these **ghost transactions** with the **highest possible accuracy**. Every percentage point of improvement in accuracy prevents significant financial losses for the company, and the current data teams have reached a dead end in finding an optimal solution.

The internal data team's theory is that the underlying pattern is not hidden in user behavior, but rather in the system's **processing infrastructure**. For this reason, several internal metadata features, which are believed to be the key to solving the mystery, have been included in this dataset.

The dataset consists of features collected from customers, including:

<i>Column Name</i>	<i>Brief Description</i>
<i>processing_batch_id</i>	Processing batch identifier
<i>transaction_amount</i>	Transaction amount
<i>user_age_days</i>	User account age in days
<i>ip_risk_score</i>	IP risk score
<i>transaction_speed_seconds</i>	How many seconds does the transaction take?
<i>user_id</i>	User identifier
<i>product_category</i>	Product category
<i>payment_method</i>	Payment method
<i>is_fraud</i>	Target column (ghost transaction)

Evaluation

Your system will be evaluated using the following formula:

Note: **The final submission file for this problem must contain exactly two columns: `user_id` and `is_fraud`.**